

1.2 RAM

RAM is short for Random Access Memory. In common parlance, RAM refers to system memory, the temporary storage area that holds all the data that the CPU is working on. All data that requires processing by the CPU and the results of the processing are first stored in the system memory.

RAM, per se, refers to the type of storage medium that allows any data stored within to be directly accessed without having to access the preceding data (the latter mode being called sequential access).

System memory is volatile, which means that the stored data is lost when the device is powered down.

1.2.1 CAS

To be able to access data from a particular storage area in RAM, the RAS and CAS procedures are needed to provide the address of the storage area. The RAM is organised as an array of storage areas. Each storage area can therefore be uniquely identified by the number of the row and the number of the column that intersect at that position. Row Access Strobe (RAS) is the process of identifying the row from which data needs to be read. This step precedes CAS. Column Access Strobe is the process of identifying the column from which data is to be accessed.

1.2.2 CAS Latency

Refers to the delay between the CAS and the arrival of the data from the RAM. It is an indicator of the speed of the memory, and is expressed in the number of clock cycles: a CAS Latency of 3 means that three clock cycles are needed after the CAS for data to be produced by the RAM.

1.2.3 DDR SDRAM

Double Data Rate SDRAM was an evolutionary improvement over SDRAM. It allowed data to be transferred twice during every cycle.